

PRACTICLE :- 11

Problem statement:-

Linux internally manages files using data structures known as inodes, which store metadata about files rather than file names. In addition, Linux provides powerful mechanisms such as input/output redirection, pipes, and process control commands to efficiently manage command execution and system resources. These concepts are fundamental for automation, scripting, system monitoring, and cyber security operations.

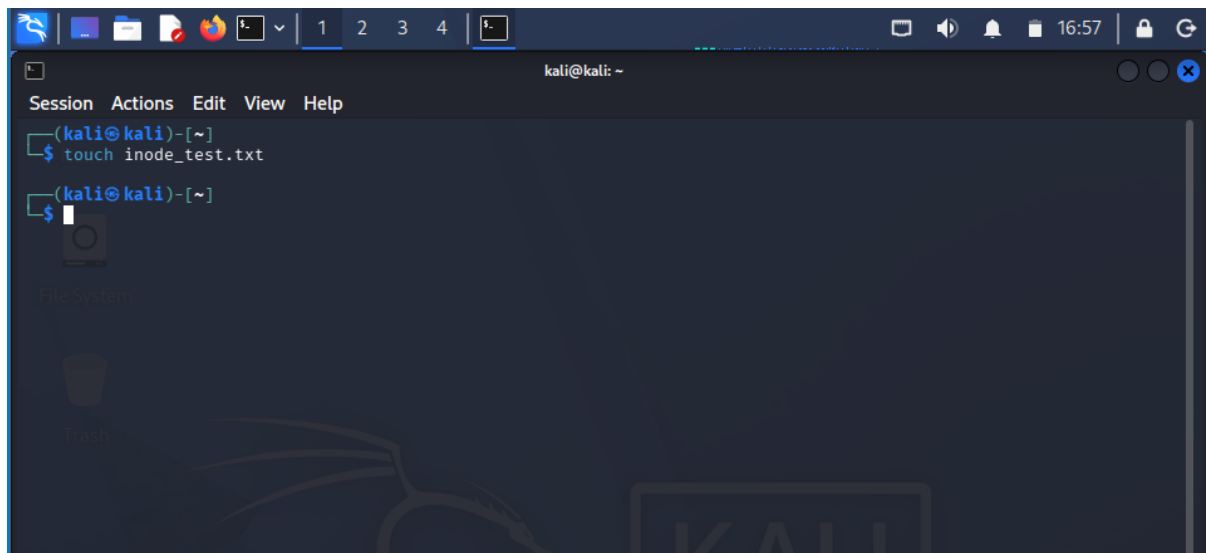
The problem is to understand the concept of inodes, practice I/O redirection and piping, and use process control commands to manage running processes in a Linux environment.

Aim :- Understanding inodes, I/O redirection, Piping and Process control commands in linux

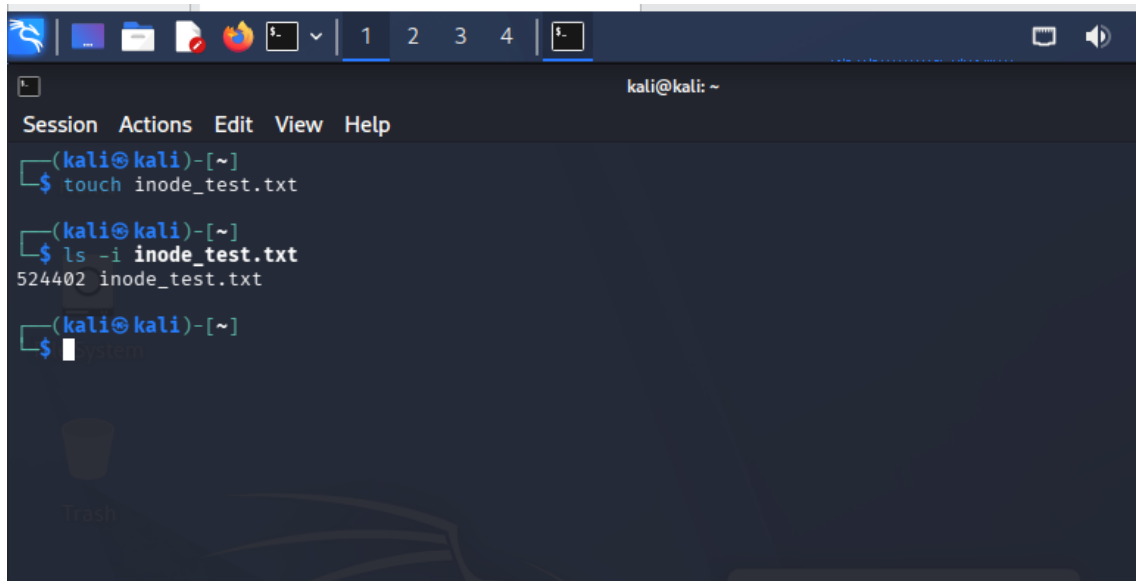
Procedure:-

A. Understanding Inodes

1. Create a test file: touch inode_test.txt

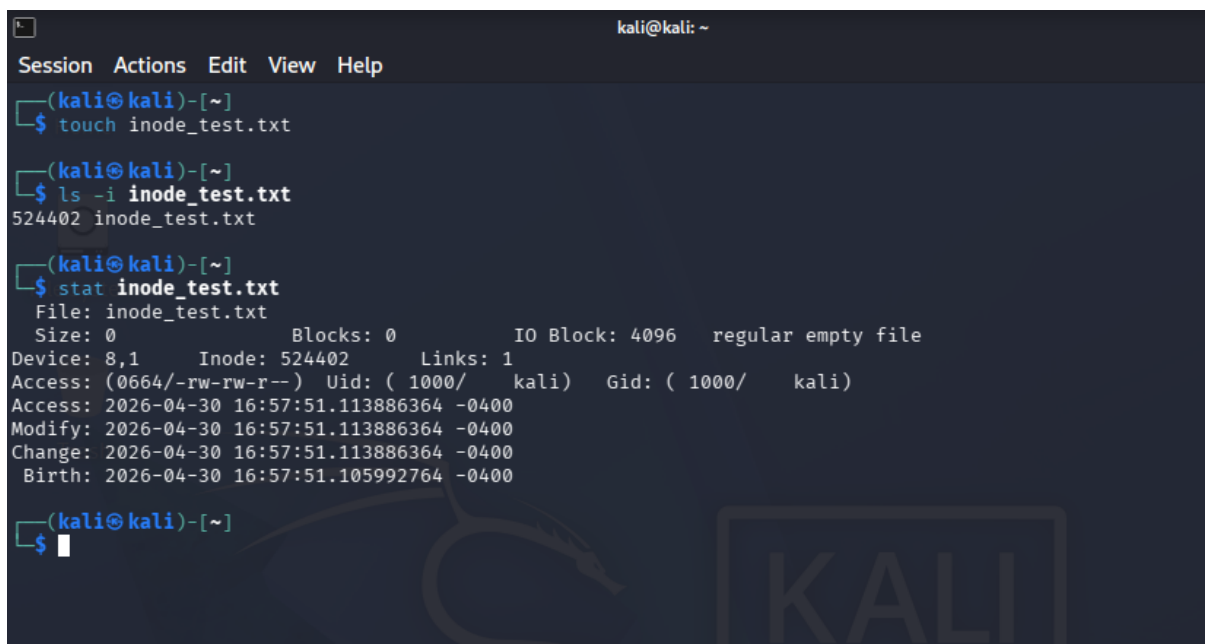
A screenshot of a terminal window titled 'kali@kali: ~'. The window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows a prompt '(kali@kali)-[~]' followed by the command '\$ touch inode_test.txt'. Below the command, there is a cursor and a small icon of a hard drive. The terminal background is dark blue with a faint 'KALI' logo at the bottom.

2. Display inode number: ls -li inode_test.txt



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ touch inode_test.txt  
(kali@kali)-[~]  
$ ls -li inode_test.txt  
524402 inode_test.txt  
(kali@kali)-[~]  
$
```

3. View inode details: stat inode_test.txt



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ touch inode_test.txt  
(kali@kali)-[~]  
$ ls -li inode_test.txt  
524402 inode_test.txt  
(kali@kali)-[~]  
$ stat inode_test.txt  
File: inode_test.txt  
Size: 0          Blocks: 0          IO Block: 4096   regular empty file  
Device: 8,1      Inode: 524402     Links: 1  
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)  
Access: 2026-04-30 16:57:51.113886364 -0400  
Modify: 2026-04-30 16:57:51.113886364 -0400  
Change: 2026-04-30 16:57:51.113886364 -0400  
Birth: 2026-04-30 16:57:51.105992764 -0400  
(kali@kali)-[~]  
$
```

B. Input / Output Redirection

4. Redirect output to a file: ls > file_list.txt

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ touch inode_test.txt  
(kali@kali)-[~]  
$ ls -li inode_test.txt  
524402 inode_test.txt  
(kali@kali)-[~]  
$ stat inode_test.txt  
File: inode_test.txt  
Size: 0          Blocks: 0          IO Block: 4096   regular empty file  
Device: 8,1      Inode: 524402      Links: 1  
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)  
Access: 2026-04-30 16:57:51.113886364 -0400  
Modify: 2026-04-30 16:57:51.113886364 -0400  
Change: 2026-04-30 16:57:51.113886364 -0400  
Birth: 2026-04-30 16:57:51.105992764 -0400  
(kali@kali)-[~]  
$ ls >file_list.txt  
(kali@kali)-[~]  
$
```

5. Append output: date >> file_list.txt

```
(kali@kali)-[~]  
$ date >> file_list.txt  
(kali@kali)-[~]  
$
```

6. Redirect input: wc -l < file_list.txt

```
(kali@kali)-[~]  
$ date >> file_list.txt  
(kali@kali)-[~]  
$ wc -l < file_list.txt  
15  
(kali@kali)-[~]  
$
```

C. Using Pipes

7. Combine commands using pipe: `ls -l | grep ".txt"`

```
(kali@kali)-[~]  
$ ls -l | grep ".txt"  
-rw-rw-r-- 1 kali kali      166 Apr 30 17:02 file_list.txt  
-rw-rw-r-- 1 kali kali       0 Apr 30 16:57 inode_test.txt  
  
(kali@kali)-[~]  
$
```

8. Count number of text files: `ls | grep ".txt" | wc -l`

```
(kali@kali)-[~]  
$ ls -l | grep ".txt"  
-rw-rw-r-- 1 kali kali      166 Apr 30 17:02 file_list.txt  
-rw-rw-r-- 1 kali kali       0 Apr 30 16:57 inode_test.txt  
  
(kali@kali)-[~]  
$ ls | grep ".txt" | wc -l  
2  
  
(kali@kali)-[~]  
$
```

D. Process Control Commands

9. Display running processes: `ps`

```
(kali@kali)-[~]  
$ ps  
  PID TTY          TIME CMD  
 77257 pts/0    00:00:02 zsh  
 82329 pts/0    00:00:00 ps  
  
(kali@kali)-[~]  
$
```

10. Monitor processes in real-time: `top`

```
(kali㉿kali)-[~]
└─$ top
top - 17:08:06 up 2:32, 3 users, load average: 0.01, 0.05, 0.05
Tasks: 237 total, 1 running, 236 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.4 us, 1.1 sy, 0.0 ni, 98.4 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 1963.0 total, 288.9 free, 899.7 used, 933.9 buff/cache
MiB Swap: 953.7 total, 953.7 free, 0.0 used, 1063.3 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
  935 root        20   0   471564 147428 85316 S   1.3   7.3   1:18.92 Xorg
    15 root        20   0         0         0     0 I   0.3   0.0   0:04.52 rcu_preempt
  585 root        20   0   253024 10472   8720 S   0.3   0.5   0:14.23 vmtoolsd
  608 root        20   0   18144   9100   7740 S   0.3   0.5   0:00.58 systemd-logind
 1729 kali        20   0 1221788 140104 91532 S   0.3   7.0   0:35.51 xfwm4
 1787 kali        20   0  432780  27636 20204 S   0.3   1.4   0:00.38 Thunar
 1846 kali        20   0  315384  67204 24808 S   0.3   3.3   0:30.96 wrapper-2.0
 1848 kali        20   0  277060  30220 22900 S   0.3   1.5   0:31.55 wrapper-2.0
 1936 root        20   0   319020 11052   8736 S   0.3   0.5   0:05.54 upowerd
 2249 kali        20   0  849860  48304 36396 S   0.3   2.4   0:19.92 vmtoolsd
 3051 kali        20   0  484104  20924 17892 S   0.3   1.0   0:00.23 xdg-desktop-por
 77242 kali        20   0 809960  68600 51804 S   0.3   3.4   0:01.93 qterminal
82714 kali        20   0 10456 5988 3820 R   0.3   0.3   0:00.07 top
     1 root        20   0   24656 15708 11364 S   0.0   0.8   0:02.11 systemd
     2 root        20   0         0         0     0 S   0.0   0.0   0:00.01 kthreadd
     3 root        20   0         0         0     0 S   0.0   0.0   0:00.00 pool_workqueue_release
     4 root        0 -20         0         0     0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
     5 root        0 -20         0         0     0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
     6 root        0 -20         0         0     0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
     7 root        0 -20         0         0     0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
     8 root        0 -20         0         0     0 I   0.0   0.0   0:00.00 kworker/R-netns
    10 root        0 -20         0         0     0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
    12 root        20   0         0         0     0 I   0.0   0.0   0:00.00 kworker/u128:0-ipv6_addrconf
    13 root        0 -20         0         0     0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
    14 root        20   0         0         0     0 S   0.0   0.0   0:00.08 ksoftirqd/0
    16 root        20   0         0         0     0 S   0.0   0.0   0:00.09 rcu_exp_par_gp_kthread_worker/1
    17 root        20   0         0         0     0 S   0.0   0.0   0:00.09 rcu_exp_gp_kthread_worker
    18 root        rt   0         0         0     0 S   0.0   0.0   0:00.08 migration/0
    19 root       -51   0         0         0     0 S   0.0   0.0   0:00.00 idle_inject/0
    20 root        20   0         0         0     0 S   0.0   0.0   0:00.00 cpuhp/0
    21 root        20   0         0         0     0 S   0.0   0.0   0:00.00 cpuhp/1
    22 root       -51   0         0         0     0 S   0.0   0.0   0:00.00 idle_inject/1
    23 root        rt   0         0         0     0 S   0.0   0.0   0:00.12 migration/1
    24 root        20   0         0         0     0 S   0.0   0.0   0:00.07 ksoftirqd/1
    26 root        0 -20         0         0     0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
```

11. Run a command in background: sleep 60 &

```
(kali㉿kali)-[~]
└─$ sleep 60 &
[1] 84919

(kali㉿kali)-[~]
└─$
```

12. List background jobs: jobs

```
(kali㉿kali)-[~]
└─$ sleep 60 &
[1] 84919

(kali㉿kali)-[~]
└─$ jobs
[1] + running  sleep 60

(kali㉿kali)-[~]
└─$
```

13. Bring job to foreground: fg

```
(srishti_m@kali)-[~]
$ sleep 60 &
[1] 15893

(srishti_m@kali)-[~]
$ jobs
[1]+  Running                  sleep 60 &

(srishti_m@kali)-[~]
$ fg
sleep 60
```

14. Terminate a process: kill

```
(srishti_m@kali)-[~]
$ sleep 60 &
[1] 15893

(srishti_m@kali)-[~]
$ jobs
[1]+  Running                  sleep 60 &

(srishti_m@kali)-[~]
$ fg
sleep 60

(srishti_m@kali)-[~]
$ kill
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]

(srishti_m@kali)-[~]
$
```

Expected Output

- Inode numbers and file metadata displayed correctly.
- Command output redirected and appended successfully.
- Pipes used to filter and process output.
- Processes monitored and controlled successfully.

Result

Inode information, input/output redirection, piping, and process control commands were successfully executed. The student gained practical understanding of Linux file system internals and process management.